Final project proposal
CS-370 Python for Data Science
Spring 2026

Project title: National Parks

Team leader: Jasmine
Other team members: Addison Thompson, Jasmine Sheeter, Jocelyn Torres, Pen Hanks

1. Data and descriptions (source, number of files, format, number of records in each file, etc.)

   New-370 data

   - National Park Visitor Use
     - NPS Visitor Use Statistics
       - Look up a specific park and then go into: summary of visitor use by month and year
     - Monthly summary of the number of visitors and what they were doing in a specific park
     - Made into 1 CSV file through Google Sheets
     - ~8000 rows
   - Visitor Spending Effects
     - Example of 2024's report
     - Economic output from parks into local economies
     - 2015-2025 for all National Parks
     - Made into 1 CSV file through Google Sheets
     - ~600 rows
   - Additional datasets were used for location data
     - Regional Designations & GPS Coordinates

2. The project goal, tentative algorithms, and approaches

   Research Question: How can this data show how National Parks economically improve the community that surrounds them?

   - Look at the economics of parks
   - See if the number of visitors aligns with the greatest economic output
   - Start with EDA to get a basic understanding, and go into more specifics with time

3. Research questions or hypotheses
    - Which parks have the most:
        - Recreation visits
        - Economic outcome
        - Jobs supported
    - Most popular seasons for visitors
    - Different visit types
    - Comparing different regions
    - Would parks be doing better economically if we ignore COVID?
    - What will the next few years look like for economic output

    Hypotheses
    - A decrease in visitors during COVID
    - A decrease in visitors during the winter/off seasons
    - Parks receive significantly more visitors during the summer months than in the winter months
    - Parks with more visitors will have a higher economic output

4. Forecast/model techniques.
    - Simple linear regression models
    - Sns plots & Matlib
    - Look at how the economic output of parks would be without COVID
        - Future predictions
    - Interactive tableau maps
        - Good visual for how economic output looks (greatest is visible)

5. New skills or knowledge need to be learned
    - Data cleaning
    - Combining data sets
        - The National Park Service provides many different data sets, and we will have to combine economic and visitation data
    - Will possibly have to learn new data visualization methods
    - Tableau!!!

6. Foreseeable roadblock or difficulties
    - Gaps in years/months in data collection
    - Combining datasets may pose issues
    - Relying on various park rangers for data collection
        - What is considered to be part of economic output → boundary area is decided with information to park rangers (we don't know what establishments are all a part of economic output)
    - Collaborating on coding may be logistically difficult

7. A rough timeline and task division (if applicable)

Week 6

- [x] ~~Finish project proposal~~
- [x] ~~Decide SSRD - decided against~~

Week 7

- [x] ~~Proposal due Feb 23~~
- [x] ~~Find out if we can use bistro data - we can't :(~~

Week 8

- [x] ~~Create a new proposal and find a new data set~~

Week 9

- [x] ~~Decide teaching topic~~
    - [x] ~~Find out how to access Tableau~~
- [x] ~~Work together on cleaning data~~

Week 10

- [x] ~~Continue cleaning~~
- [x] ~~Figure out the best way to have a shared workspace?~~

Week 11

- [x] ~~Start looking at questions, split up individually for specific questions~~
- [x] ~~Start researching our research topic~~
- [x] ~~Have preliminary research ready to present~~

Week 12

- [x] ~~Continue working on specific questions~~
- [x] ~~Preliminary NPS presentation~~

Week 13

- [x] ~~Come together to review our findings~~
- [x] ~~Finalize our code and visualization~~

Week 14

- [x] ~~Have the slideshow made~~

4/7 Project preliminary presentation (Progress check)
5/5 Final project presentation

8. Other information
    - Opted out of ssrd